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MATHEMATICS

0580/22

Paper 2 Non-calculator (Extended)

February/March 2026

2 hours

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages.

List of formulas

Area, A , of triangle, base b , height h .

$$A = \frac{1}{2}bh$$

Area, A , of circle of radius r .

$$A = \pi r^2$$

Circumference, C , of circle of radius r .

$$C = 2\pi r$$

Curved surface area, A , of cylinder of radius r , height h .

$$A = 2\pi rh$$

Curved surface area, A , of cone of radius r , sloping edge l .

$$A = \pi rl$$

Surface area, A , of sphere of radius r .

$$A = 4\pi r^2$$

Volume, V , of prism, cross-sectional area A , length l .

$$V = Al$$

Volume, V , of pyramid, base area A , height h .

$$V = \frac{1}{3}Ah$$

Volume, V , of cylinder of radius r , height h .

$$V = \pi r^2 h$$

Volume, V , of cone of radius r , height h .

$$V = \frac{1}{3}\pi r^2 h$$

Volume, V , of sphere of radius r .

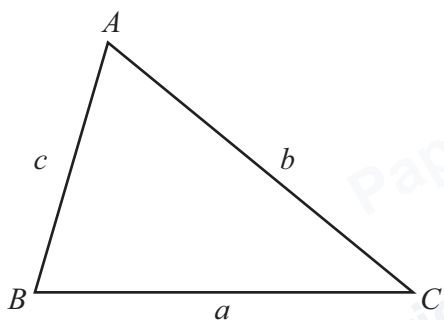
$$V = \frac{4}{3}\pi r^3$$

For the equation

$$ax^2 + bx + c = 0, \text{ where } a \neq 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For the triangle shown,



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}ab \sin C$$

Calculators must **not** be used in this paper.

1 Work out.

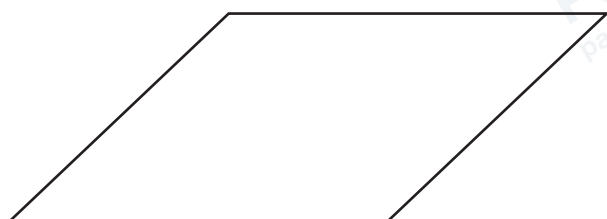
(a) 0.7^2

..... [1]

(b) $2 \div 0.04$

..... [1]

2



The diagram shows a parallelogram.

Complete the statement.

A parallelogram has lines of symmetry

and rotational symmetry of order

[1]



3

-50	-11	9	15	27	50	83
-----	-----	---	----	----	----	----

(a) From this list of numbers, write down

(i) a prime number

..... [1]

(ii) a cube number

..... [1]

(iii) two numbers with a product of 550.

..... and [1]

(b) Rishwan writes each of these numbers on a card.

-50	-11	9	15	27	50	83
-----	-----	---	----	----	----	----

Rishwan picks a card at random and replaces it.

(i) Find the probability that he picks a natural number.

..... [1]

(ii) He repeats this 140 times.

Find the expected number of times that Rishwan picks a natural number.

..... [1]

4 Simplify.

$$a \times 2a + 3a$$

..... [1]



- 5 A theatre has 3 types of seating: stalls, circle or private box.
The cost of a ticket for each type of seating is shown in the table.

type of seating	cost of a ticket
stalls	\$30
circle	\$24
private box	\$72

- (a) Find the ratio of the cost of the tickets stalls : circle : private box.
Give your answer in its simplest form.

..... : : [2]

- (b) Arshad buys tickets for the theatre on a website.
The website charges a fee of 5% of the cost of each ticket.
Arshad buys 2 tickets in the stalls and 3 tickets in the circle.

Work out the total cost of Arshad's tickets.

\$ [3]

- (c) There are 1200 seats in the stalls.
One evening, $\frac{5}{6}$ of the tickets for seats in the stalls are sold.
On this evening, the number of tickets sold for seats is in the ratio stalls : circle = 10 : 7.
Find the number of tickets sold for seats in the circle.

..... [3]

- 6 Nisha plays basketball.

Here are some statistics about the number of points she has scored in her last 4 games:

- The mean is 10.
- The median is 13.
- The range is 12.

Find the number of points she scored in each of her last 4 games.

..... [3]

- 7 Line L has equation $y = 2$.

(a) Find the equation of the line perpendicular to line L that passes through the point $(-1, 5)$.

..... [1]

(b) Find the coordinates of the point where line L intersects the line $y = 20 - 6x$.

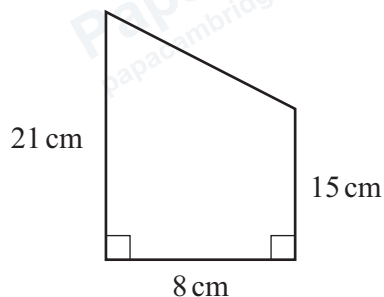
(..... ,) [2]

- 8 Find the integer values of x that satisfy $-3 \leq x < 1$.

..... [2]



9



NOT TO SCALE

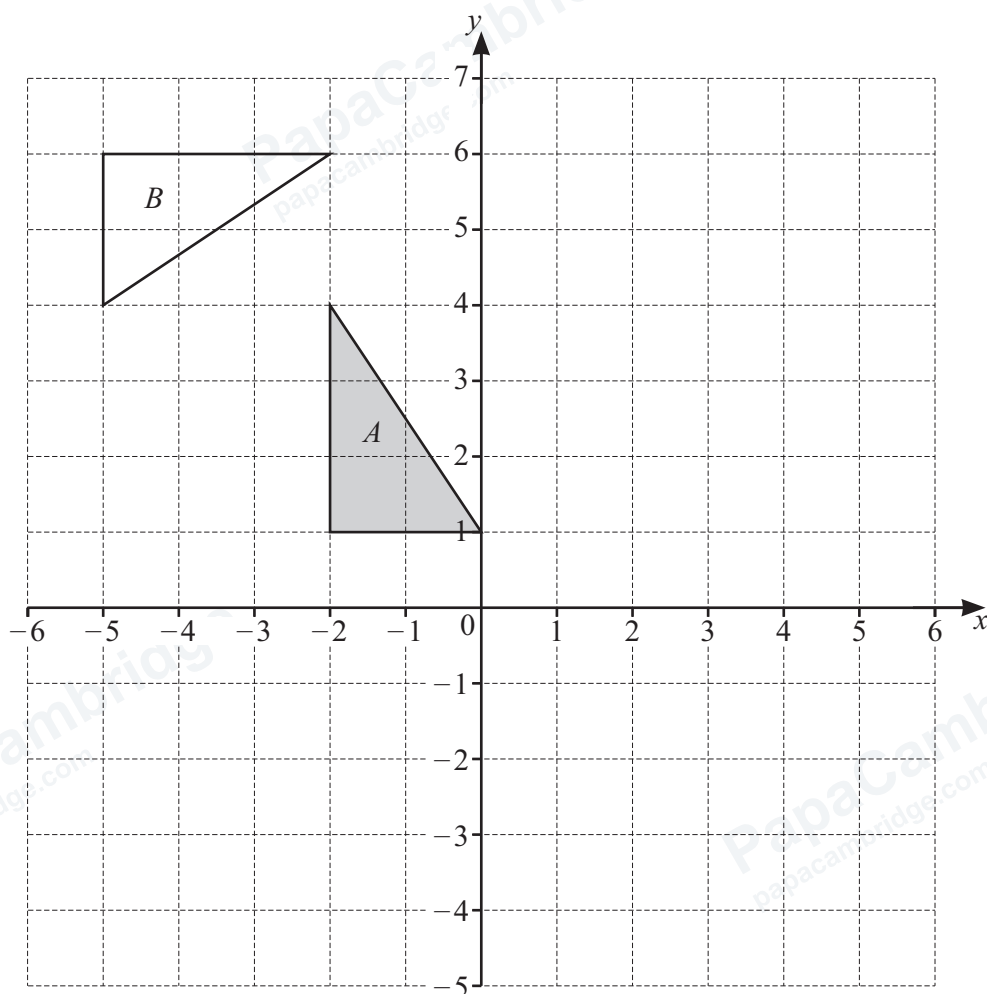
Work out the perimeter of the trapezium.

..... cm [3]

- 10 The bearing of A from B is $(2x + 50)^\circ$.
The bearing of B from A is 060° .

Find the value of x .

$x =$ [3]



(a) On the grid, draw the image of

(i) triangle A after a translation by the vector $\begin{pmatrix} -3 \\ -5 \end{pmatrix}$ [2]

(ii) triangle A after a reflection in the line $y = x$. [2]

(b) Describe fully the **single** transformation that maps triangle A onto triangle B .

..... [3]

12 (a) Write 230 000 in standard form.

..... [1]

(b) Work out $(4.2 \times 10^{32}) - (4.2 \times 10^{31})$.

Give your answer in standard form.

..... [2]

(c) Work out $(80 \times 10^{50})^{\frac{1}{3}}$.

Give your answer in standard form.

..... [2]

13 $\mathcal{C} = \{x : x \text{ is an integer where } 40 < x < 50\}$

$P = \{x : x \text{ is a square number}\}$

$Q = \{x : x \text{ is an odd number}\}$

$R = \{x : x \text{ is a multiple of 3}\}$

(a) Use set notation to complete the statement.

$P \cap R = \dots\dots\dots [1]$

(b) Find $n(P \cap Q \cap R)'$.

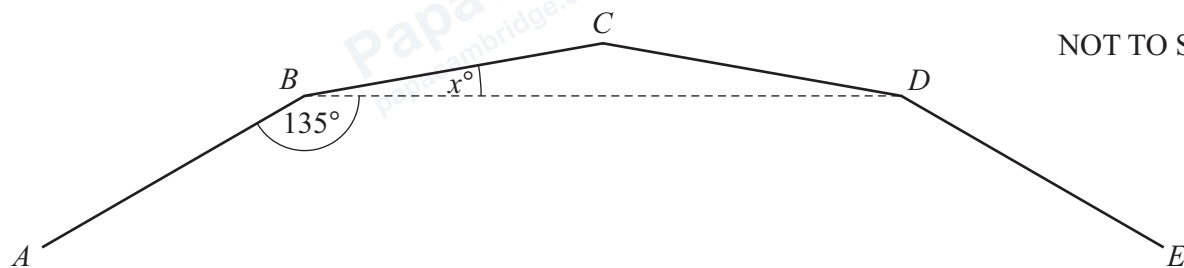
..... [1]



14



10



The diagram shows part of a regular polygon.
Angle $ABD = 135^\circ$ and angle $CBD = x^\circ$.

By finding the value of x , work out the number of sides of this polygon.

..... [4]

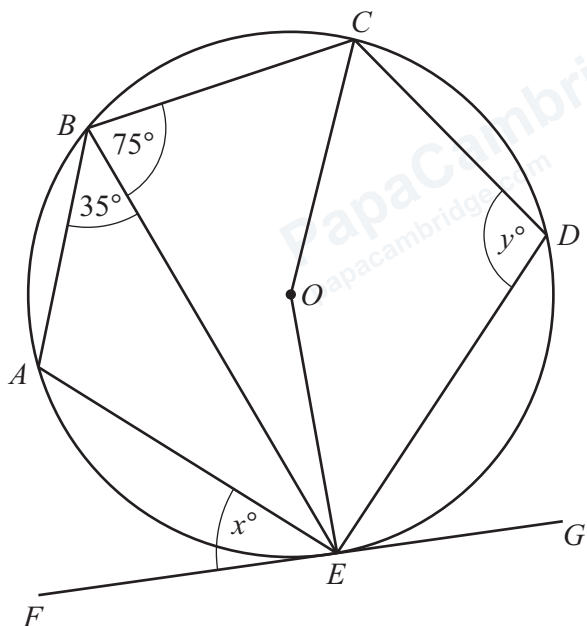
15 Simplify.

$$\sqrt{40} + \sqrt{160}$$

..... [2]



16



NOT TO SCALE

A, B, C, D and E lie on a circle, centre O .
 FG is a tangent to the circle at E .
 Angle $ABE = 35^\circ$ and angle $CBE = 75^\circ$.

(a) Find the value of x and the value of y .

$x =$

$y =$

[2]

(b) Complete this statement with the geometrical reason.

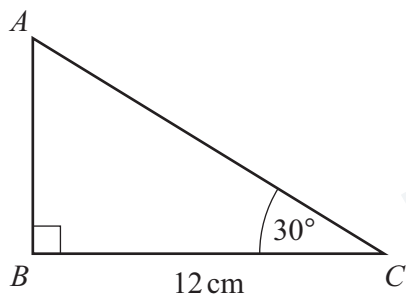
Obtuse angle $EOC = 150^\circ$ because

..... [1]

(c) The radius of the circle is 6 cm.

Work out the area of the major sector EOC .
 Give your answer in terms of π in its simplest form.

..... cm^2 [3]



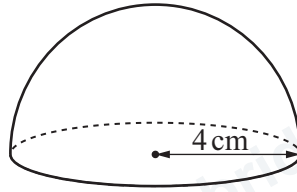
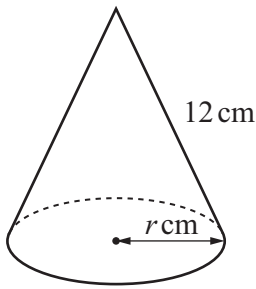
NOT TO SCALE

- (a) Find the length of AC .
Give your answer in the form $a\sqrt{b}$.

..... cm [4]

- (b) Find the shortest distance from B to AC .

..... cm [3]



NOT TO SCALE

The diagram shows a cone and a hemisphere.

The cone has radius r cm and sloping edge 12 cm.

The hemisphere has radius 4 cm.

The **total** surface area of the cone is equal to the **total** surface area of the hemisphere.

(a) Show that $r^2 + 12r - 48 = 0$.

[3]

(b) Find the exact value of the radius of the cone.
Give your answer as a surd in its simplest form.

$r = \dots\dots\dots$ [4]

19 Point A has coordinates $(-4, 3)$ and point B has coordinates $(4, 7)$.

(a) Find the vector $\overrightarrow{BA}$.

$$\overrightarrow{BA} = \begin{pmatrix} \\ \end{pmatrix} \quad [2]$$

(b) Point C has coordinates $(x, 7)$ and $|\overrightarrow{AC}| = 4\sqrt{10}$.

Find the two possible values of x .

$$x = \dots\dots\dots \text{ or } x = \dots\dots\dots \quad [4]$$

(c) Point D lies on the line AB .
 $AD : DB = 3 : 1$.

Find the equation of the line perpendicular to AB that passes through point D .
 Give your answer in the form $y = mx + c$.

$$y = \dots\dots\dots \quad [5]$$



20 (a) Simplify.

$$(27x^9)^{-\frac{2}{3}}$$

..... [3]

(b) Solve.

$$7^x = 49^{(1-2x)}$$

$x =$ [3]

21 Simplify.

$$\frac{2x^2 + 3x - 20}{ax + 4a - 3bx - 12b}$$

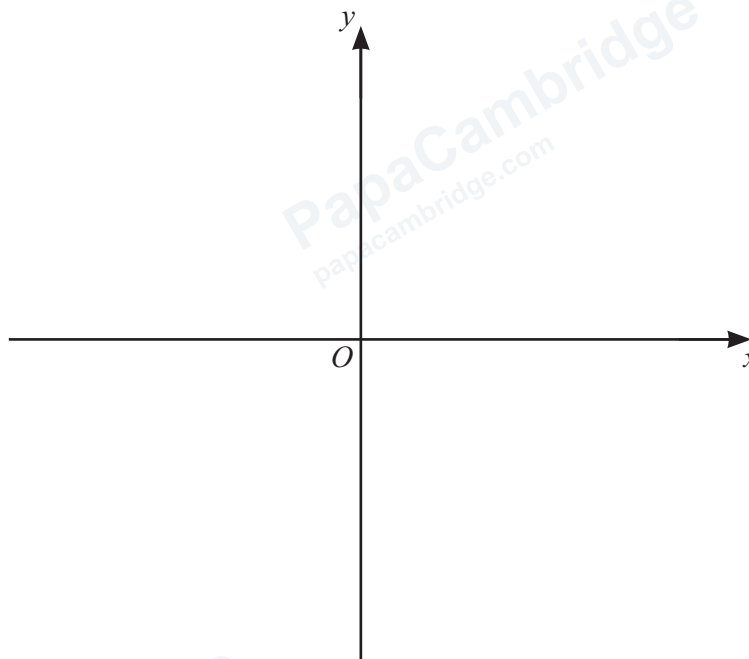
..... [5]

Question 22 is on the back page.

22 (a) Expand and simplify $(x+5)(x-3)^2$.

..... [3]

(b) On the diagram, sketch the graph of $y = (x+5)(x-3)^2$, indicating the values where the graph crosses the axes.



[4]

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